

# A Radial-Twist Counterexample to a Coarse-Sum Coordinate Problem

Automated Math Discovery Smoke Test

June 8, 2026

## Source and Open Question

This packet concerns Problem 5.7 from:

Bruno de Mendonca Braga, *Asymptotic structure and coarse Lipschitz geometry of Banach spaces*, arXiv:1604.08661; Studia Mathematica 237 (2017), 71–97.

The source location is page 20 of the arXiv PDF. In the parsed source, the problem appears at lines 980–984 of `source.tex`.

$$d(cC)^{-\delta} - Ck^{-\delta} \leq 2K Ck^{\delta/42-\delta/12} C^{-\delta} + K Ck^{\delta/42-\delta} + 3\delta$$

for all  $k \in \mathbb{N}$ , and all  $\delta \in (0, 1)$ . As  $q_2 > p > 1$ , by letting  $k \rightarrow \infty$  and  $\delta \rightarrow 0$ , we get a contradiction.  $\square$

If  $T = (T_1, T_2) : X \rightarrow Y_1 \oplus Y_2$  is a linear isomorphic embedding, then either  $T_1 : X \rightarrow Y_1$  or  $T_2 : X \rightarrow Y_2$  is not strictly singular, i.e.,  $T_i : X_0 \rightarrow Y_i$  is a linear isomorphic embedding, for some infinite dimensional subspace  $X_0 \subset X$ , and some  $i \in \{1, 2\}$ . Is there an analog of this result for coarse Lipschitz embeddings? Precisely, we ask the following.

**Problem 5.7.** Let  $X, Y_1$  and  $Y_2$  be Banach spaces and consider a coarse Lipschitz embedding  $f = (f_1, f_2) : X \rightarrow Y_1 \oplus Y_2$ . Is there an infinite dimensional subspace  $X_0 \subset X$  such that either  $f_1 : X_0 \rightarrow Y_1$  or  $f_2 : X_0 \rightarrow Y_2$  is a coarse Lipschitz embedding?

We can now prove Theorem 1.6, which will be essential in the proof of Theorem 1.7.

*Proof of Theorem 1.6.* Say  $m \in \{1, \dots, n-1\}$  is such that  $p \in (p_m, p_{m+1})$  (the other cases have analogous proofs). Then  $(T^{p_{m+1}} \oplus \dots \oplus T^{p_n})_\epsilon$  is reflexive (see

Problem 5.7 asks whether, for every coarse Lipschitz embedding

$$f = (f_1, f_2) : X \rightarrow Y_1 \oplus Y_2,$$

there is an infinite-dimensional subspace  $X_0 \subset X$  such that either  $f_1|_{X_0} : X_0 \rightarrow Y_1$  or  $f_2|_{X_0} : X_0 \rightarrow Y_2$  is a coarse Lipschitz embedding.

## Answer

The answer is negative. We construct a bi-Lipschitz embedding

$$F = (F_1, F_2) : \ell_2 \rightarrow \ell_2 \oplus_2 \ell_2$$

such that neither coordinate is a coarse embedding on any infinite-dimensional subspace of  $\ell_2$ .

## Construction

Let  $H = \ell_2$  be a real Hilbert space. Define

$$d_4(t) = \text{dist}(t, 4\mathbb{Z}), \quad \varphi(t) = \frac{\pi}{4} \min\{d_4(t), 2\} \quad (t \geq 0),$$

and

$$\theta(r) = \varphi(\log(1+r)) \quad (r \geq 0).$$

Then  $\varphi$  is  $\pi/4$ -Lipschitz, takes values in  $[0, \pi/2]$ , and

$$\theta(e^{4k} - 1) = 0, \quad \theta(e^{4k+2} - 1) = \frac{\pi}{2}$$

for all  $k \in \mathbb{N} \cup \{0\}$ .

For  $x \in H$ , set

$$F(x) = (\cos(\theta(\|x\|))x, \sin(\theta(\|x\|))x) \in H \oplus_2 H.$$

Write  $F = (F_1, F_2)$  for its two coordinate maps.

## The Full Map is Bi-Lipschitz

**Theorem 1.** *For all  $x, y \in H$ ,*

$$\frac{1}{\sqrt{2}} \|x - y\| \leq \|F(x) - F(y)\|_{H \oplus_2 H} \leq \left(1 + \frac{\pi}{4}\right) \|x - y\|.$$

*In particular,  $F$  is a coarse Lipschitz embedding.*

*Proof.* For  $t \in [0, \pi/2]$ , define the isometry

$$J_t : H \rightarrow H \oplus_2 H, \quad J_t z = (\cos t z, \sin t z).$$

Then  $F(x) = J_{\theta(\|x\|)}x$ .

Let  $r = \|x\|$  and  $s = \|y\|$ . If both  $x$  and  $y$  are nonzero, write  $u = x/r$ ,  $v = y/s$ ,  $a = \langle u, v \rangle$ , and  $c = \cos(\theta(r) - \theta(s))$ . Since  $\theta$  takes values in  $[0, \pi/2]$ , we have  $c \in [0, 1]$ . A direct calculation gives

$$\|F(x) - F(y)\|^2 = r^2 + s^2 - 2rsca,$$

whereas

$$\|x - y\|^2 = r^2 + s^2 - 2rsa.$$

If  $a \geq 0$ , then  $ca \leq a$ , so

$$\|F(x) - F(y)\| \geq \|x - y\|.$$

If  $a < 0$ , then

$$\|F(x) - F(y)\|^2 \geq r^2 + s^2 \geq \frac{1}{2} \|x - y\|^2.$$

The same lower estimate is immediate if one of  $x, y$  is zero. Hence

$$\|F(x) - F(y)\| \geq 2^{-1/2} \|x - y\|.$$

For the upper estimate, assume without loss of generality that  $r \geq s$ . Then

$$\begin{aligned}\|F(x) - F(y)\| &\leq \|J_{\theta(r)}(x - y)\| + \|(J_{\theta(r)} - J_{\theta(s)})y\| \\ &\leq \|x - y\| + s|\theta(r) - \theta(s)|.\end{aligned}$$

Since  $\varphi$  is  $\pi/4$ -Lipschitz,

$$|\theta(r) - \theta(s)| \leq \frac{\pi}{4} \log \frac{1+r}{1+s}.$$

Using  $\log(1+u) \leq u$  with  $u = (r-s)/(1+s)$ , we get

$$s \log \frac{1+r}{1+s} \leq \frac{s}{1+s}(r-s) \leq r-s \leq \|x-y\|.$$

Therefore

$$\|F(x) - F(y)\| \leq \left(1 + \frac{\pi}{4}\right) \|x - y\|.$$

□

## The Coordinate Maps Fail on Every Infinite-Dimensional Subspace

**Theorem 2.** *Let  $X_0 \subset H$  be any nonzero linear subspace. Then neither  $F_1|_{X_0}$  nor  $F_2|_{X_0}$  is a coarse embedding. Consequently, the conclusion asked for in Problem 5.7 fails for every infinite-dimensional subspace  $X_0 \subset H$ .*

*Proof.* Choose  $u \in X_0$  with  $\|u\| = 1$ .

For  $F_1$ , set

$$R_k = e^{4k+2} - 1, \quad x_k = R_k u.$$

Then  $\theta(R_k) = \pi/2$ , so

$$F_1(x_k) = F_1(-x_k) = 0.$$

However,

$$\|x_k - (-x_k)\| = 2R_k \rightarrow \infty.$$

Thus the compression modulus of  $F_1|_{X_0}$  is zero at arbitrarily large scales, so  $F_1|_{X_0}$  is not a coarse embedding.

For  $F_2$ , set

$$S_k = e^{4k} - 1, \quad y_k = S_k u.$$

Then  $\theta(S_k) = 0$ , so

$$F_2(y_k) = F_2(-y_k) = 0,$$

while

$$\|y_k - (-y_k)\| = 2S_k \rightarrow \infty.$$

Therefore  $F_2|_{X_0}$  is not a coarse embedding either.

□

## Conclusion

Taking  $X = Y_1 = Y_2 = \ell_2$  and using the map  $F = (F_1, F_2)$  above gives a coarse Lipschitz embedding into the two-coordinate direct sum, but no coordinate map is a coarse Lipschitz embedding on any infinite-dimensional subspace. This gives a negative answer to Problem 5.7.

**Remark 1.** *The proof was written for the Hilbert two-sum  $Y_1 \oplus_2 Y_2$ . If the direct sum in the problem is read with the  $\ell_1$  norm, the same example still works: on  $Y_1 \times Y_2$  the  $\ell_1$  and  $\ell_2$  norms are equivalent, so the full map remains a coarse Lipschitz embedding, and the coordinate-collapse argument is unchanged.*

## Verification Notes

The verification script in the `code` folder samples the same construction in finite-dimensional Hilbert space. It checks that the sampled distance ratios lie within the proved constants

$$2^{-1/2} \quad \text{and} \quad 1 + \pi/4,$$

and verifies the exact coordinate collapses at the radii  $e^{4k+2} - 1$  and  $e^{4k} - 1$ .

## Novelty Check

I searched for the exact text of Problem 5.7 and for later papers using the phrases “coarse Lipschitz embedding”, “ $Y_1 \oplus Y_2$ ”, and “strictly singular”. The search found the source paper and related coarse-Lipschitz geometry papers, but no later paper explicitly answering this problem. This packet should still be treated as a candidate result until a human checks the surrounding literature more thoroughly.

# Human Review: radial-twist sum-coordinate counterexample

Original automatic proof: `solution_packet.pdf`

Checked date: 10 June 2026

Status: Manually checked.

Verdict: Appears correct.

## Human Comments

The solution appears to be correct. The central idea is to encode the information of a vector  $x$  into two coordinates, with the split depending on the radius  $\|x\|$ . Concretely, the construction has the form

$$F(x) = (\cos(\theta(\|x\|))x, \sin(\theta(\|x\|))x).$$

Thus the full map  $F = (F_1, F_2)$  still remembers  $x$  in the two-coordinate sum, but each individual coordinate alternately loses all information on carefully chosen radial shells.

The alternation is governed by the norm of  $x$ . On one sequence of radii,  $\theta(\|x\|) = 0$ , so the second coordinate vanishes:

$$F_2(x) = 0.$$

On another sequence of radii,  $\theta(\|x\|) = \pi/2$ , so the first coordinate vanishes:

$$F_1(x) = 0.$$

The full map remains a coarse Lipschitz embedding because the information is not destroyed; it is merely moved between the two coordinates as the radius changes.

## Strength of the Result

The packet proves something stronger than what is needed for the stated problem. It does not merely show that there is no infinite-dimensional subspace on which one coordinate becomes a coarse Lipschitz embedding. Rather, for every nonzero linear subspace  $X_0$ , neither

$$F_1|_{X_0} \quad \text{nor} \quad F_2|_{X_0}$$

is a coarse embedding.

Indeed, after choosing any unit vector  $u \in X_0$ , the points  $R_k u$  and  $-R_k u$  become arbitrarily far apart in  $X_0$ , while one of the coordinate maps sends both to the same point. The same argument applies to the other coordinate along the alternating sequence of radii. Thus the failure already occurs on every one-dimensional subspace, not merely on infinite-dimensional ones.

## Short Version

The proof works by a radial twist. The vector  $x$  is split between two coordinates according to  $\|x\|$ , and the active coordinate alternates at larger and larger scales. The pair  $F = (F_1, F_2)$  is still a coarse Lipschitz embedding, but each separate coordinate collapses arbitrarily distant pairs in every nonzero subspace.

## Literature and Follow-Up

We did not find this result stated in the literature during the present review. Since the surrounding problem is specific and the construction is quite natural once seen, it may be good to reach out to the author to verify whether the example is already known and to get his take on its significance.

## Review Outcome

Archive this packet as an apparently correct negative solution to the coarse-sum coordinate problem, with the additional note that the construction is stronger than the stated obstruction: no coordinate map works even after restriction to a nonzero finite-dimensional subspace.